

Werdegang einer Hartmetallschere



Spaltteil

Slit part



Schmiedeteil

Forging blank



Stanzteil

Forging part
with ring eye



Rohteil-Stanzteil
Ring gelocht

Forging part
with ring eye punched



Innen- und
Außenring
schleifen

Grind ring inside
and outside



Bohrung,
Halm schleifen,
Schluss und
HM-Einlage ausfräsen

Drill hole grind branch,
mill out carbide insert



HM-Einlage
aufschweißen

Weld TC



Schluss und Innenblatt
schleifen

Grind end and inner blade



Scheren paaren,
Seriennummer
anbringen, härten,
Außenblatt schleifen

Match scissors, attach serial number,
harden grind outer blade



Hartrichten

Check alignment



Halm polieren

Polish branch



Maulbreite schleifen,
demontieren,
gleitschleifen,
elektropolieren

Grind jaw width,
disassemble,
vibratory grinding,
electropolishing



Innen- und Außenblatt
polieren,
Schneidkante scharf
schleifen

Grind inner and outer blade,
sharpen cutting edge



Montieren,
Fertigrichten,
Spitze ausformen

Assemble and adjust screw
shape the tip



Hochglanzpolieren,
vergolden

High mirror polishing,
gold coating



Mattieren

Scotch surface



Laserbeschriftung,
Endprüfung

Laser etching,
final inspection



AITIN Beschichtung

AITIN coating

The Manufacturing Process of a Tungsten-Carbide Surgical Scissor

Technical Overview

Precision Steel Blank: Surgical-grade stainless steel blanks are selected for metallurgical consistency, corrosion resistance, and dimensional stability.

Forging: Blanks are hot-forged to establish grain flow, structural strength, and baseline geometry.

Ring Formation: Finger rings are punched and formed to precise ergonomic dimensions.

Profile Refinement: Forged parts are machined to achieve final scissor geometry and balance.

Surface Grinding: Internal and external surfaces are ground to ensure alignment and smooth operation.

Pivot & Blade Machining: Pivot holes and blade seats are machined to micron-level tolerances.

Carbide Bonding: Tungsten-carbide inserts are permanently bonded using controlled high-temperature processes.

Carbide Edge Grinding: Carbide cutting edges are ground to exact cutting geometry.

Blade Matching: Opposing blades are matched and ground to ensure full-length shear contact.

Heat Treatment: Components are heat-treated to achieve final hardness and fatigue resistance.

Shank Polishing: Handles are polished to remove surface imperfections and improve handling.

Edge Refinement: Cutting edges are sharpened and refined for performance consistency.

Assembly & Tensioning: Scissors are assembled and hand-adjusted for precise blade tension.

Final Finishing: High-gloss or matte finishes are applied based on product specification.

Laser Marking: Permanent laser markings are applied for traceability and identification.

Advanced Coatings: Optional PVD or AlTiN coatings are applied to enhance wear resistance and visibility.